THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 8: **Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Three numbers in AP have sum 45 and product 2640 (with $d > 0$). The largest number is: 1

(a) 22 **(b)** 29 **(c)** 23 **(d)** 15

2. A town's population is 5000 and grows by 10% each year. After 3 years it is: 1

(a) 6755 **(b)** 6705 **(c)** 6655 **(d)** 6605

3. The first term of a GP is 5 and the 9th term is 1280 (with $r > 0$). The common ratio is: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 1

4. The nth term of an AP is $t_n = 2n + 7$. The sum of its first 20 terms is: 1

(a) 560 **(b)** 558 **(c)** 562 **(d)** 580

5. The 4th term of an AP is 3 and the 9th term is 33. The 13th term is: 1

(a) 57 **(b)** 51 **(c)** 69 **(d)** 63

6. The sum of the first n terms of a sequence is $S_n = 1n^2 + 6n$. Its 15th term is: 1

(a) 35 **(b)** 34 **(c)** 37 **(d)** 36

7. The sum of all integers from 1 to 300 that are divisible by 13 is: 1

(a) 3598 **(b)** 3601 **(c)** 3575 **(d)** 3588

8. The sum of the first 20 terms of an AP is 640 and its first term is -6. The common difference is: 1

(a) 6 **(b)** 3 **(c)** 5 **(d)** 4

9. **Assertion (A):** The 10th term of the AP $3, 7, 11, \ldots$ is 39. 1
Reason (R): The nth term of an AP is $t_n = a + (n-1)d$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** In an AP with first term 7 and common difference -2, the 5th term is -1. 1
Reason (R): The common difference of an AP can be negative.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If $S_n = 5n^2 + 1n$ is the sum of the first n terms of a sequence, find its first term and its 8th term. 2

12. The sum of the first 18 terms of an AP is 297 and the first term is -9. Find d and the 18th term. 2

13. Which term of the AP $14, 18, 22, \ldots$ is the first term to exceed 111? 2

14. If $S_n = 4n^2 + 7n$ is the sum of the first n terms of a sequence, find its first term and its 15th term. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The sum of the 3rd and 7th terms of an AP is 64 and its 10th term is 62. Find the AP's first term and the sum of its first 13 terms. 3

16. Find the sum of all natural numbers between 10 and 180 that leave remainder 5 when divided by 6. 3

17. Three positive numbers are in GP. Their product is 13824 and the sum of the first and third is 60. Find the numbers. 3

OR

The sum of the 6th and 12th terms of an AP is 84 and its 4th term is 17. Find the AP's first term and the sum of its first 25 terms. 3

18. Find the sum of all natural numbers between 40 and 260 that leave remainder 3 when divided by 8. 3

OR

Three positive numbers are in GP. Their product is 27000 and the sum of the first and third is 156. Find the numbers. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A school launches a news bulletin. In stage 1, one student knows the news; in each later stage every student who knows it tells 3 new students, so the number of new students informed in successive stages is $1, 3, 9, 27, \ldots$ 5

(i) Find the common ratio.

(ii) Find the number of new students informed in stage 5.

(iii) Write the nth term.

(iv) In which stage does the number first exceed 500?

20. A contractor pays a penalty for delay: ₹250 on day 1, ₹290 on day 2, ₹330 on day 3 and so on. If the work is 22 days late, find the penalty on the last day and the total penalty. 5

OR

A ball is dropped from 162 m and rebounds to 2/3 of the previous height. Show that the rebound heights form a GP and find the 3rd rebound height. 5

END OF QUESTION PAPER — Total: 40 Marks